GA2 - Temă seminar 2

1. $A = \begin{pmatrix} 1 & 3 & 0 & -2 \\ 2 & 1 & -1 & 3 \\ 1 & 2 & 3 & -1 \\ -1 & 4 & 0 & 1 \end{pmatrix}$

$$\det A = \begin{vmatrix} 1 & 3 & 0 & -2 \\ 2 & 1 & -1 & 3 \\ 1 & 2 & 3 & -1 \\ -1 & 4 & 0 & 1 \end{vmatrix} \underset{\substack{L_2 = L_2 + \frac{3}{2} L_1 \\ L_3 = L_3 - \frac{1}{2} L_1 \\ L_4 = L_4 + \frac{1}{2} L_1}}{=\!=} \begin{vmatrix} 1 & 3 & 0 & -2 \\ \frac{7}{2} & \frac{11}{2} & -1 & 0 \\ \frac{1}{2} & \frac{1}{2} & 3 & 0 \\ -\frac{1}{2} & \frac{11}{2} & 0 & 0 \end{vmatrix} =$$

$$\underset{\text{dezv } C_4}{=\!=} (-1)^{1+4} \cdot (-2) \begin{vmatrix} \frac{7}{2} & \frac{11}{2} & -1 \\ \frac{1}{2} & \frac{1}{2} & 3 \\ -\frac{1}{2} & \frac{11}{2} & 0 \end{vmatrix} = 2\left(0 - \frac{33}{4} - \frac{11}{4} - \frac{1}{4} - \frac{231}{4} - 0\right)$$

$$= 2 \cdot \left(-\frac{276}{4}\right) = -\frac{276}{2} = -138$$

2. $A_3(1,2) = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$

$$\left(\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \underset{L_2 \Leftrightarrow L_1}{\Rightarrow} \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right)$$

$$\underset{\substack{L_2 = L_2 - 2L_1 \\ L_3 = L_3 - L_1}}{\Rightarrow} \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & -3 & -1 & 1 & -2 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 \end{array}\right) \underset{\substack{L_3 \Leftrightarrow L_2 \\ L_2 = -L_2}}{\Rightarrow} \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & -3 & -1 & 1 & -2 & 0 \end{array}\right)$$

$$\Rightarrow \quad L_1 = L_1 - 2L_2, \quad L_3 = L_3 + 3L_1 \qquad \cancel{\begin{pmatrix} 1 & 0 & 3 \\ 0 & -3 & 3 \\ 0 & 0 & 4 \end{pmatrix}} \quad \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & -1 & 2 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & 0 & -4 & 1 & 1 & -3 \end{array}\right) \Rightarrow$$

$$\Rightarrow \quad L_4 = -\frac{L_4}{4} \quad \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & -1 & 2 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & +\frac{3}{4} \end{array}\right) \Rightarrow \begin{matrix} L_2 = L_2 + L_3 \\ L_1 = L_1 - 3L_3 \end{matrix} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ 0 & 1 & 0 & -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{array}\right)$$

$$\Rightarrow \quad A_3(1,2)^{-1} = \begin{pmatrix} \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{pmatrix}$$

3. $$\Delta_1 = \begin{vmatrix} x & a & a & a \\ a & x & a & a \\ a & a & x & a \\ a & a & a & x \end{vmatrix} = 0 \quad \underset{L_1 = L_1 + L_2 + L_3 + L_4}{\Longrightarrow}$$

$$\Delta_1 = \begin{vmatrix} x+3a & x+3a & x+3a & x+3a \\ a & x & a & a \\ a & a & x & a \\ a & a & a & x \end{vmatrix} =$$

$$= (x+3a) \begin{vmatrix} 1 & 1 & 1 & 1 \\ a & x & a & a \\ a & a & x & a \\ a & a & a & x \end{vmatrix} \underset{\substack{C_2 = C_2 - C_1 \\ C_3 = C_3 - C_1 \\ C_4 = C_4 - C_1}}{=\!=} (x+3a) \begin{vmatrix} 1 & 0 & 0 & 0 \\ a & x-a & 0 & 0 \\ a & 0 & x-a & 0 \\ a & 0 & 0 & x-a \end{vmatrix}$$

$$\underset{\text{dezv } L_1}{=} (x+3a) \cdot 1 \cdot (-1)^{1+1} \begin{vmatrix} x-a & 0 & 0 \\ 0 & x-a & 0 \\ 0 & 0 & x-a \end{vmatrix} = (x+3a)(x-a)^3$$

$$(x+3a)(x-a)^3=0 \Rightarrow \begin{cases} x_1=-3a \\ x_2=x_3=x_4=a \end{cases}$$

4. $$\Delta_n = \begin{vmatrix} -1 & a & a & \dots & a \\ a & -1 & a & \dots & a \\ a & a & -1 & \dots & a \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a & a & a & \dots & -1 \end{vmatrix} \overset{\substack{L_2=L_2-L_1 \\ L_3=L_3-L_1 \\ \vdots \\ L_n=L_n-L_1}}{=\!=} \begin{vmatrix} -1 & a & a & \dots & a \\ a+1 & -(a+1) & 0 & \dots & 0 \\ a+1 & 0 & -(a+1) & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a+1 & 0 & 0 & \dots & -(a+1) \end{vmatrix} =$$

$$\underset{C_1=C_1+C_2+\dots+C_n}{=\!=} \begin{vmatrix} (n-1)a-1 & a & a & \dots & a \\ 0 & -(a+1) & 0 & \dots & 0 \\ 0 & 0 & -(a+1) & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & -(a+1) \end{vmatrix} \underset{\text{dezv } C_1}{=\!=}$$

$$((n-1)a-1)\cdot(-1)^{1+1} \begin{vmatrix} -(a+1) & 0 & \dots & 0 \\ 0 & -(a+1) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -(a+1) \end{vmatrix} =$$

$$= ((n-1)a-1)\cdot(-1)^{n-1}\cdot(a+1)^{n-1}$$

5. $$\Delta_2 = \begin{vmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_1 \\ x_3 & x_1 & x_2 \end{vmatrix}$$

x_1, x_2, x_3 răd ec $x^3-2x^2+2x+17=0$

$s_1 = x_1+x_2+x_3 = 2$

$s_2 = x_1x_2+x_1x_3+x_2x_3 = 2$

$S_1 = x_1^2+x_2^2+x_3^2 = s_1^2 - 2s_2 = 4-4=0$

$$\Delta_2 \underset{C_1 = C_1 + C_2 + C_3}{=} (x_1+x_2+x_3)\begin{vmatrix} 1 & x_2 & x_3 \\ 1 & x_3 & x_1 \\ 1 & x_1 & x_2 \end{vmatrix} = (x_1+x_2+x_3)(x_2x_3 + x_1x_2 + x_1x_3 - x_3^2 - x_2^2 - x_1^2)$$

$$\Delta_2 = \underbrace{(x_1+x_2+x_3)}_{\sigma_1}(\underbrace{x_1x_2 + x_1x_3 + x_2x_3}_{\sigma_2} - \underbrace{(x_1^2+x_2^2+x_3^2)}_{S_1})$$

$$\Delta_2 = 2\cdot(2-0) = 4$$

6. a) $\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ yz & xz & xy \end{vmatrix} \overset{?}{=} (x-y)(y-z)(z-x)(xy+xz+yz)$

$$\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ yz & xz & xy \end{vmatrix} \underset{\substack{C_2 = C_2 - C_1 \\ C_3 = C_3 - C_1}}{=} \begin{vmatrix} x & y-x & z-x \\ x^2 & y^2-x^2 & z^2-x^2 \\ yz & z(x-y) & y(x-z) \end{vmatrix} =$$

$$= (y-x)(z-x)\cdot\begin{vmatrix} x & 1 & 1 \\ x^2 & y+x & z+x \\ yz & -z & -y \end{vmatrix} \underset{C_1 = C_1 - xC_2}{=} (y-x)(z-x)\begin{vmatrix} 0 & 1 & 1 \\ -xy & y+x & z+x \\ z(y+x) & -z & -y \end{vmatrix}$$

$$\underset{C_2 = C_2 - C_3}{=} (y-x)(z-x)\begin{vmatrix} 0 & 0 & 1 \\ -xy & y-z & z+x \\ z(y+x) & -(z-y) & -y \end{vmatrix} = (y-x)(z-x)(y-z)\begin{vmatrix} 0 & 0 & 1 \\ -xy & 1 & z+x \\ z(y-x) & -1 & -y \end{vmatrix}$$

$$\underset{\text{dezv } L_1}{=} (y-x)(z-x)(y-z)\cdot 1\cdot(-1)^{1+1}\begin{vmatrix} -xy & 1 \\ z(y+x) & +1 \end{vmatrix} =$$

$$= -(x-y)(y-z)(z-x)(-xy - zy - zx) = (x-y)(y-z)(z-x)(xy + zy + zx)$$

b) $\begin{vmatrix} a+b & b+c & c+a \\ a^2+b^2 & b^2+c^2 & c^2+a^2 \\ a^3+b^3 & b^3+c^3 & c^3+a^3 \end{vmatrix} = 2abc(a-b)(b-c)(c-a)$

$$\begin{vmatrix} a+b & b+c & c+a \\ a^2+b^2 & b^2+c^2 & c^2+a^2 \\ a^3+b^3 & b^3+c^3 & c^3+a^3 \end{vmatrix} \underset{\substack{c_2=c_2-c_1 \\ c_3=c_3-c_1}}{=} \begin{vmatrix} a+b & c-a & c-b \\ a^2+b^2 & c^2-a^2 & c^2-b^2 \\ a^3+b^3 & c^3-a^3 & c^3-b^3 \end{vmatrix} =$$

$$=(c-a)(c-b)\begin{vmatrix} a+b & 1 & 1 \\ a^2+b^2 & c+a & c+b \\ a^3+b^3 & c^2+ca+a^2 & c^2+cb+b^2 \end{vmatrix} \underset{\substack{c_1=c_1-ac_2- \\ -bc_3}}{=} \begin{vmatrix} 0 \\ -c(a+b) \\ -ca(a+c)-cb(b+c) \end{vmatrix}$$

$$= (c-a)(c-b)\begin{vmatrix} 0 & 1 & 1 \\ -c(a+b) & c+a & c+b \\ -ca(a+c)-cb(b+c) & c^2+ca+a^2 & \cancel{c}^2+cb+b^2 \end{vmatrix} \underset{c_2=c_2-c_3}{=}$$

$$=-c(c-a)(c-b)\begin{vmatrix} 0 & 0 & 1 \\ +(a+b) & a-b & c+b \\ a(a+c)+b(b+c) & c(a-b)+a^2-b^2 & c^2+cb+b^2 \end{vmatrix} \underset{dezv\, l_1}{=}$$

$$=-c(c-a)(c-b)\cdot 1\cdot(-1)^{1+1}\begin{vmatrix} +(a+b) & a-b \\ a(a+c)+b(b+c) & c(a-b)+a^2-b^2 \end{vmatrix} =$$

$$=-c(a-b)(c-a)(c-b)\begin{vmatrix} +(a+b) & 1 \\ a(a+c)+b(b+c) & a+b+c \end{vmatrix}$$

$$=-c(a-b)(c-a)(c-b)(a^2+b^2+2ab+ac+bc-a^2-b^2-ac-bc)$$

$$=2abc(a-b)(b-c)(c-a)$$

7. a) $\Delta_3 = \begin{vmatrix} 1 & 3 & 0 \\ 0 & 1 & 3 \\ 3 & 0 & 1 \end{vmatrix} = 1 + 3^3 + 0 - 0 - 0 - 0 = 28$

$$\Delta_4 = \begin{vmatrix} 1 & 3 & 0 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 3 \\ 3 & 0 & 0 & 1 \end{vmatrix} \underset{L_4 = L_4 - 3L_1}{=} \begin{vmatrix} 1 & 3 & 0 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & -9 & 0 & 1 \end{vmatrix} \underset{\text{dezv } c_1}{=}$$

$$= 1 \cdot (-1)^{1+1} \begin{vmatrix} 1 & 3 & 0 \\ 0 & 1 & 3 \\ -9 & 0 & 1 \end{vmatrix} = 1 - 3^4 + 0 - 0 - 0 - 0 = 1 - 81 = -80$$

b) $$\Delta_n = \begin{vmatrix} 1 & 3 & 0 & \dots & 0 & 0 \\ 0 & 1 & 3 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 3 \\ 3 & 0 & 0 & \dots & 0 & 1 \end{vmatrix} \underset{\text{dezv } c_1}{=}$$

$$= 1 \cdot (-1)^{1+1} \begin{vmatrix} 1 & 3 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{vmatrix} + 3 \cdot (-1)^{n+1} \begin{vmatrix} 3 & 0 & \dots & 0 \\ 1 & 3 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 3 \end{vmatrix}$$

$$= 1 \cdot (-1)^2 \cdot 1^{n-1} + (-1)^{n+1} \cdot 3^n$$

$$= 1 + (-1)^{n+1} \, 3^n$$

8. a) $\Delta_n = \begin{vmatrix} 4 & 3 & 0 & \dots & 0 \\ 1 & 4 & 3 & \dots & 0 \\ 0 & 1 & 4 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 4 \end{vmatrix} \underset{\text{dezv } L_1}{=}$

$= 4 \cdot (-1)^{1+1} \underbrace{\begin{vmatrix} 4 & 3 & \dots & 0 \\ 1 & 4 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 4 \end{vmatrix}}_{=\Delta_{n-1}} + 3 \cdot (-1)^{1+2} \begin{vmatrix} 1 & 3 & 0 & \dots & 0 \\ 0 & 4 & 3 & \dots & 0 \\ 0 & 1 & 4 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & 3 \\ 0 & 0 & 0 & \dots 1 & 4 \end{vmatrix} =$

$= 4\Delta_{n-1} - 3 \cdot 1 \cdot (-1)^{1+1} \underbrace{\begin{vmatrix} 4 & 3 & \dots & 0 \\ 1 & 4 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 4 \end{vmatrix}}_{=\Delta_{n-2}} = 4\Delta_{n-1} - 3\Delta_{n-2}$ $(\forall)\, n \geq 3$, $n \in \mathbb{N}$

b) $\Delta_n - 4\Delta_{n-1} + 3\Delta_{n-2} = 0$

$r^2 - 4r + 3 = 0$ $\begin{cases} r_1 = 1 \\ r_2 = 3 \end{cases}$ $\Rightarrow$ $\Delta_n = A r_1^n + B r_2^n$

$\Delta_n = A \cdot 1^n + B \cdot 3^n$

$= 3^n B + A$; A, B const

$\Delta_3 = \begin{vmatrix} 4 & 3 & 0 \\ 1 & 4 & 3 \\ 0 & 1 & 4 \end{vmatrix} = 64 - 12 - 12 = 40$

$\Delta_4 = \begin{vmatrix} 4 & 3 & 0 & 0 \\ 1 & 4 & 3 & 0 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 1 & 4 \end{vmatrix} \underset{L_1 = L_1 - 4L_2}{=} \begin{vmatrix} 0 & -13 & -12 & 0 \\ 1 & 4 & 3 & 0 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 1 & 4 \end{vmatrix} \underset{\text{dezv } c_1}{=}$

$$\Delta_4 = 1 \cdot (-1)^{1+2} \begin{vmatrix} -13 & -12 & 0 \\ 1 & 4 & 3 \\ 0 & 1 & 4 \end{vmatrix} = -(-208 - 0 + 0 - 0 + 39 + 48) = 121$$

$$\begin{cases} 27B + A = 40 \\ 81B + A = 121 \end{cases} \Rightarrow 54B = 81 \Rightarrow B = \frac{81}{54} = \cancel{\frac{7}{6}}\ \frac{3}{2} \Rightarrow A = -\frac{1}{2}$$

$$\Rightarrow \Delta_n = \frac{3}{2} \cdot 3^n - \frac{1}{2} = \frac{3^{n+1} - 1}{2} \quad (\forall)\ n \geq 3,\ n \in \mathbb{N}$$